

KARTHIK SARVAN

Rust & Embedded Systems Engineer

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ABOUT ME

Electronics and Communication

Engineering student with a keen interest in computers, software, and embedded systems. Eager to delve into the intersection of technology and innovation, I am committed to learning and contributing to these dynamic fields. I specialize in writing safe, concurrent systems in Rust, exploring low-level kernel development, and bridging the gap between embedded hardware and AI-driven architectures.

WORK EXPERIENCE

Aegion Dynamics

Sept 2024 – Present
Visakhapatnam, India

Embedded Intern

Gaining in-depth hands-on experience in the design, development, and optimization of embedded solutions. Focused on microcontroller-based system design with a strong emphasis on real-time hardware-software integration. Designed and analyzed electronic circuits, interfaced multiple sensors and actuators, and developed efficient low-level firmware using C and Rust ensuring reliability and performance.

Spotmies

Sept 2024 – Present
Visakhapatnam, India

MERN Stack Developer

Working as a MERN Stack Developer at REAIDY, a product-based company focused on interview preparation. My role involves developing Reaidy AI and guiding my team members. Contributing heavily to a platform designed for specialized interview practice.

Spotmies

June 2023 – Dec 2023
Visakhapatnam, India

React Intern

Completed an internship as a Web Application Developer specializing in the MERN stack. Gained production-level expertise in React, MongoDB, Git, Node.js, Express, and other related technologies while collaborating with cross-functional teams and clients.

FEATURED PROJECTS

Pratibim

Rust, AI, Embedded
Software Platform

Core Developer

Conceptualized and developed an innovative AI-driven software platform customized specifically for electronics engineers and hardware designers. Optimized mission-critical computational modules in Rust to parse, manage, and scale complex underlying electronic system structures with memory safety and zero-cost abstractions.

Mimic

Robotics, C/C++
Hardware Lab

Firmware Engineer

Engineered an advanced robotics tracking framework focused entirely on ultra-low-latency motion capture processing, spatial mapping, and micro-controller instruction feedback loops. Wrote efficient hardware drivers and high-performance controller firmware to eliminate transport lag.

EDUCATION

Chaitanya Engineering College

2023 – Present
Pursuing B.Tech in Electronics & Communication

Gov. Polytechnic Pendurti

Completed
Diploma in Electronics & Communication

SKILLS

Languages

Rust, C/C++, JavaScript, Python (Basic)

Frameworks

MERN Stack, React, Node.js

Systems & Hardware

Embedded Firmware, Linux Kernel, PCB Design, Git